

Roll No

AU-704 (GS)**B.E. VII Semester**

Examination, December 2017

Grading System (GS)**Combustion and Heat Transfer***Time : Three Hours**Maximum Marks : 70*

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.
 iii) Use of HMT data book permitted.

1. a) Explain the phenomenon of Knocking in SI engines. Compare it with CI engine knock. 7
 b) Explain the various stages of combustion in CI engine. 7
2. a) What are the various types of combustion chamber used in SI engine? Explain them briefly. 7
 b) Explain the various factors affecting the delay period in CI engine. 7
3. a) Discuss in brief the methods of measuring the fuel consumption in IC engine. 7
 b) Derive the expression of heat transfer through a piston crown. 7
4. a) Define thermal conductivity. How do the thermal conductivity of liquids and gases vary with temperature. 7
 b) The wall of a furnace is constructed from 15cm thick fire brick having constant thermal conductivity of 1.7 W/mk. The two sides of the wall are maintained at 1400K and 1150K, respectively. What is the rate of heat loss through the wall which is 50cm x 3m on a side. 7

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5. a) Define critical thickness of insulation for pipe geometry. 4
 b) Calculate the amount of energy required to solder together two very long pieces of bare copper wire 1.5mm in diameter with solder that melts at 190°C. The wires are positioned vertically in air at 20°C. Assume that the heat transfer coefficient on the wire surface is 20W/m² °C and thermal conductivity of wire alloy is 330W/m °C. 10
6. a) Define Laminar and Turbulent flow. 4
 b) Air at 20°C and 1.013 bar flows over a flat plate at 40m/s. The plate is 1m long and is maintained at 60°C. Assuming unit depth, calculate the heat transfer from the plate. Use the following correlations: 10

$$Nu_L = (Pr)^{0.33} [(Re_L)^{0.8} - 850]$$
7. a) Define the terms absorptivity, reflectivity and transmissivity of radiation. 4
 b) Calculate the following for an industrial furnace in the form of a black body and emitting radiation at 2500°C. 10
 i) Monochromatic emissive power at 1.2µm length,
 ii) Wavelength at which the emission is maximum,
 iii) Maximum emissive power,
 iv) Total emissive power and
 v) Total emissive power of the furnace if it is assumed as a real surface with emissivity equal to 0.9.
8. Write short notes on any four of the following: 14
 a) Thermal stress in piston
 b) Use of air box method in IC engine testing
 c) Modes of basic laws of heat transfer
 d) Dielectric heating
 e) Efficiency and effectiveness of fin
 f) Stefan Boltzmann law
 g) Kirchhoff's law
